

Software Requirements Specification for Annapurna

Version 1.0

Prepared by Project Team 30

Name	PRN
Amol Gavit	250240320013
Meenakshi Lokhande	250240320061
Shilpa Pandey	250240520079
Pallavi Sahay	250240320072
Suchi Tripathi	250240320117

Centre for Development of Advanced Computing

Raintree Marg, Near Bharati Vidyapeeth, Opp. Kharghar Railway Station, Sector 7,
CBD Belapur, Navi Mumbai - 400 614 - Maharashtra (India)

Phone: +91-22-27565303

Fax: +91-22-2756-0004

Contents

1	Introduction	2
1.1	Purpose	2
1.2	Document Conventions	2
1.3	Intended Audience and Reading Suggestions	2
1.4	Project Scope	2
1.5	References	2
2	Overall Description	3
2.1	Product Perspective	3
2.2	Product Features	3
2.3	User Classes and Characteristics	3
2.4	Operating Environment	3
2.5	Design and Implementation Constraints	4
2.6	User Documentation	4
2.7	Assumptions and Dependencies	4
3	System Features	4
3.1	Customer Functionalities	4
3.1.1	Description and Priority	4
3.1.2	Stimulus/Response Sequences	4
3.1.3	Functional Requirements	4
3.2	Vendor Functionalities	5
3.2.1	Description and Priority	5
3.2.2	Stimulus/Response Sequences	5
3.2.3	Functional Requirements	5
3.3	Admin Functionalities	5
3.3.1	Description and Priority	5
3.3.2	Stimulus/Response Sequences	5
3.3.3	Functional Requirements	5
4	External Interface Requirements	6
4.1	User Interfaces	6
4.2	Hardware Interfaces	6
4.3	Software Interfaces	6
4.4	Communications Interfaces	6
5	Other Nonfunctional Requirements	6
5.1	Performance Requirements	6
5.2	Safety Requirements	6
5.3	Security Requirements	7
5.4	Software Quality Attributes	7
6	Other Requirements	

1 Introduction

1.1 Purpose

This **Software Requirements Specification (SRS)** outlines the requirements for **Annapurna**, an online tiffin service platform connecting home-based food vendors with customers, including students and office workers. The goal is to provide healthy, homemade food and a way for vendors to earn from home. **Version 1.0** covers the core modules for **Customer**, **Vendor**, and **Admin**.

1.2 Document Conventions

- Follows the **IEEE SRS template**.
- A **Glossary** is provided in Appendix A for technical terms.

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- **Developers**: For development reference.
- **Testers**: For creating test cases.
- **Stakeholders**: To validate functionality.
- **Trainers & Documentation Writers**: For preparing training materials.

1.4 Project Scope

The platform enables:

- **Customers** to view, order, and pay for tiffin services.
- **Vendors** to maintain tiffin plans and track sales.
- **Admins** to approve vendors, monitor activity, and handle payments/disputes.

It supports **real-time dashboards** and **secure access** for all users.

1.5 References

- <https://reactjs.org/docs>
- <https://spring.io>
- <https://hibernate.org>

2 Overall Description

2.1 Product Perspective

Annapurna is a web-based platform comprising:

- Frontend **React**.
- **Spring Boot and ASP.NET** REST backend.
- **MySQL and MS SQL Server** database.

It can run standalone or be scaled to cloud. It is not part of another system.

2.2 Product Features

- **Customer** and **Vendor** registration.
- Tiffin plans with **search/filter** functionality.
- **Secure** order and payment processing.
- Order history and reviews.
- **Vendor** dashboard.
- **Admin** analytics panel.
- **Role-based** access control.

2.3 User Classes and Characteristics

- **Anonymous**: Browse menu.
- **Customer**: Order, review, pay.
- **Vendor**: Manage menu, view sales.
- **Admin**: Full access and analytics.

2.4 Operating Environment

- **Web browsers**: Chrome, Firefox, Edge.
- **Frontend**: ReactJS.
- **Backend**: Spring Boot and ASP.NET.
- **Database**: MySQL and MS SQL Server.
- **OS**: Windows, Linux, or Cloud.

2.5 Design and Implementation Constraints

- Must use **Java/Spring Boot** and **.NET**.
- **MySQL** required.
- **Role-based access control** enforced.

2.6 User Documentation

- Online user guide.
- **Admin** manual (PDF).
- Tutorials and demo videos.
- FAQ section.

2.7 Assumptions and Dependencies

- Users have **internet access**.
- **Third-party APIs** (e.g., Payment Gateway) are available.
- Future versions may depend on **location APIs**.

3 System Features

3.1 Customer Functionalities

3.1.1 *Description and Priority*

Enables **Customers** to register, place orders, cancel within 5 minutes, pay online, and review orders.

Priority: High

3.1.2 *Stimulus/Response Sequences*

- User registers → Confirmation email sent.
- Places order → Payment triggered → Order placed.
- Cancels within 5 minutes → Refund initiated.

3.1.3 *Functional Requirements*

1. **REQ-1:** Customers must register and log in.
2. **REQ-2:** Can browse tiffin menus and plans.
3. **REQ-3:** Can place or cancel orders within 5 minutes.
4. **REQ-4:** Can pay securely.
5. **REQ-5:** View order history.
6. **REQ-6:** Leave ratings/reviews.

3.2 Vendor Functionalities

3.2.1 Description and Priority

Enables **Vendors** to manage menu, accept/reject orders, track sales, and view feedback.

Priority: High

3.2.2 Stimulus/Response Sequences

- Vendor logs in → Adds/updates tiffin items.
- Order placed → Notification to vendor → Vendor accepts/rejects.

3.2.3 Functional Requirements

1. **REQ-7:** Vendor registration and login.
2. **REQ-8:** Manage menu items.
3. **REQ-9:** View order analytics.
4. **REQ-10:** Accept/reject orders.
5. **REQ-11:** View customer reviews.

3.3 Admin Functionalities

3.3.1 Description and Priority

Enables **Admins** to approve vendors, monitor orders, payments, and disputes.

Priority: High

3.3.2 Stimulus/Response Sequences

- Vendor registers → Admin approves.
- Complaint raised → Admin investigates and resolves.

3.3.3 Functional Requirements

1. **REQ-12:** Admin login.
2. **REQ-13:** Approve vendors and tiffin listings.
3. **REQ-14:** Track sales and transactions.
4. **REQ-15:** Manage user accounts and disputes.

4 External Interface Requirements

4.1 User Interfaces

- **Responsive UI** using HTML, CSS, and Bootstrap.
- **Login**, dashboard, and menu management screens.
- **Error and success alerts**.
- **Order tracking** panel.

4.2 Hardware Interfaces

- No special hardware required.
- Standard **browser** with **internet access**.

4.3 Software Interfaces

- **MySQL** database.
- **Payment gateway API** (Razorpay or Stripe).
- **SMTP server** for emails.
- **JWT-based** authentication interface.

4.4 Communications Interfaces

- **HTTPS** for all communication.
- **RESTful APIs** using HTTP.
- **Email** for order confirmation and updates.
- **Future**: SMS/Push notifications.

5 Other Nonfunctional Requirements

5.1 Performance Requirements

- Support **100+ concurrent users**.
- **Response time** < 2 seconds.
- **Load balanced** on server.

5.2 Safety Requirements

- **Secure payment flow**.
- **Regular backups**.
- **Data recovery** in case of crash.

5.3 Security Requirements

- **JWT authentication.**
- **Role-based authorization.**
- **SSL encryption.**
- **Auto logout** on inactivity.
- **Encrypted passwords** using BCrypt.

5.4 Software Quality Attributes

- **Availability:** 99.99% uptime.
- **Reliability:** Daily backups.
- **Maintainability:** Modular codebase.
- **Portability:** Deployable on any OS or cloud.
- **Usability:** Simple and responsive UI.

6 Other Requirements

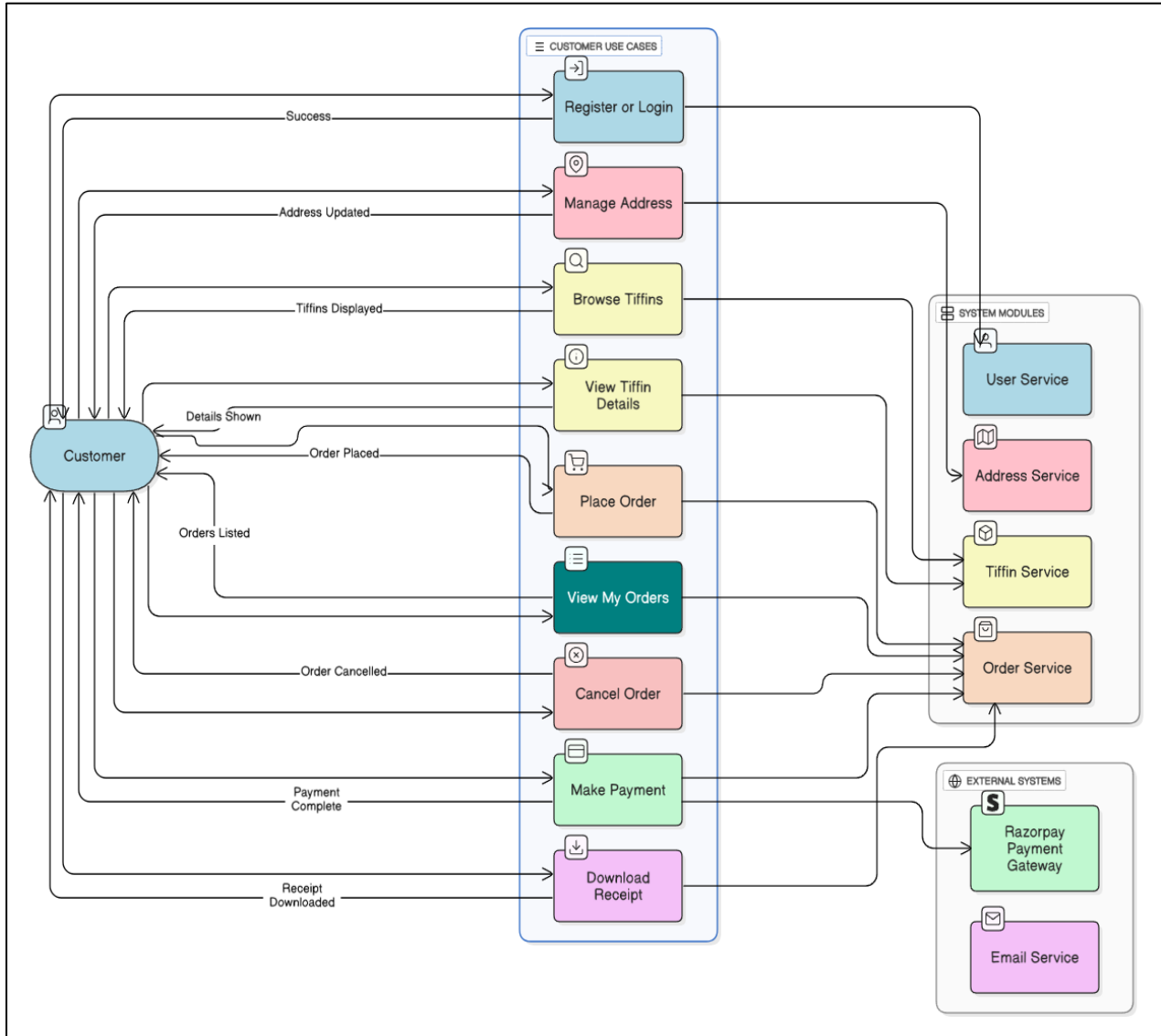
- **GDPR compliance** for customer data.
- **Multi-language support** in future versions.
- **Vendor KYC verification.**
- **Graph/chart dashboard** for analytics.

Appendix A: Glossary

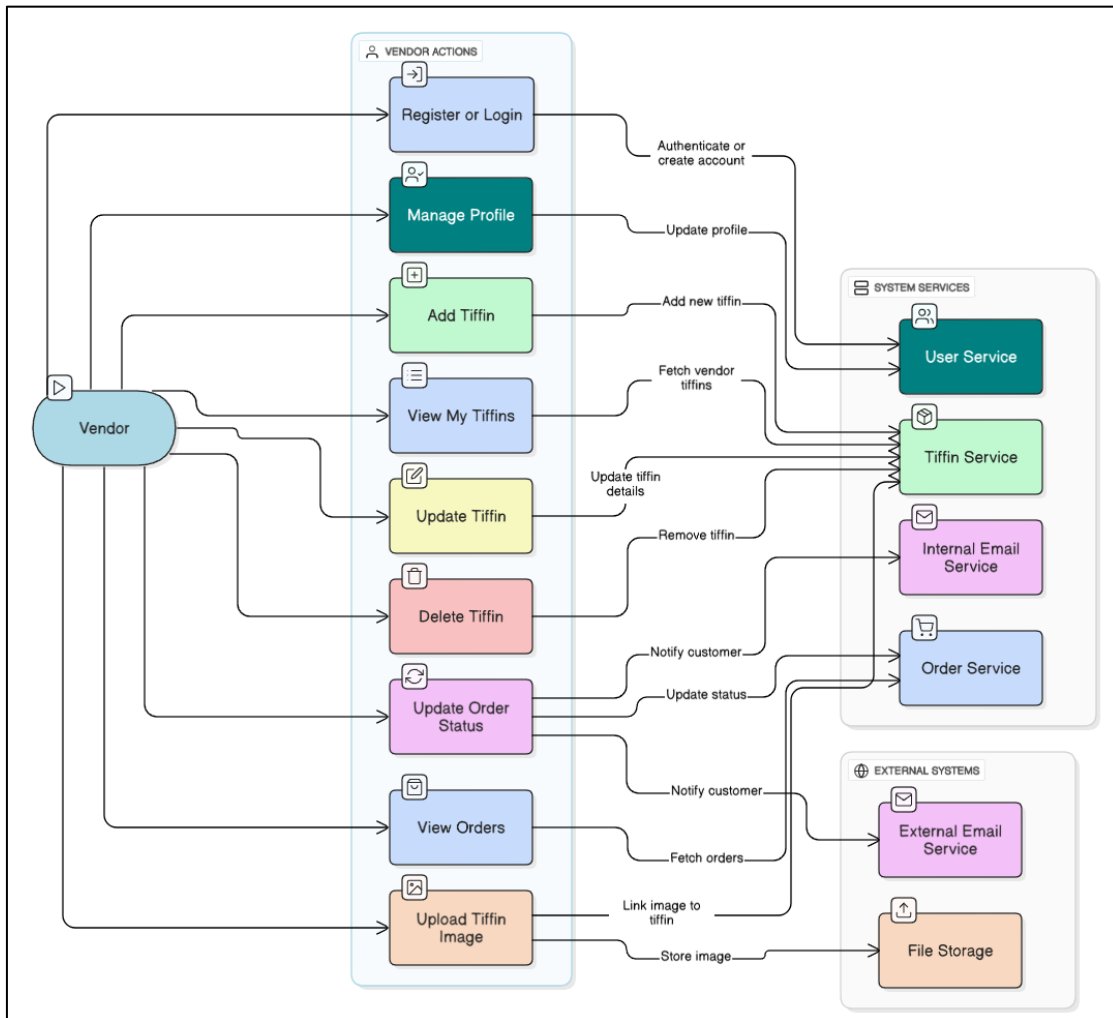
- **JWT:** JSON Web Token.
- **CRUD:** Create, Read, Update, Delete.
- **UI:** User Interface.
- **SSL:** Secure Sockets Layer.
- **DFD:** Data Flow Diagram.
- **KYC:** Know Your Customer.

Appendix B: Analysis Models

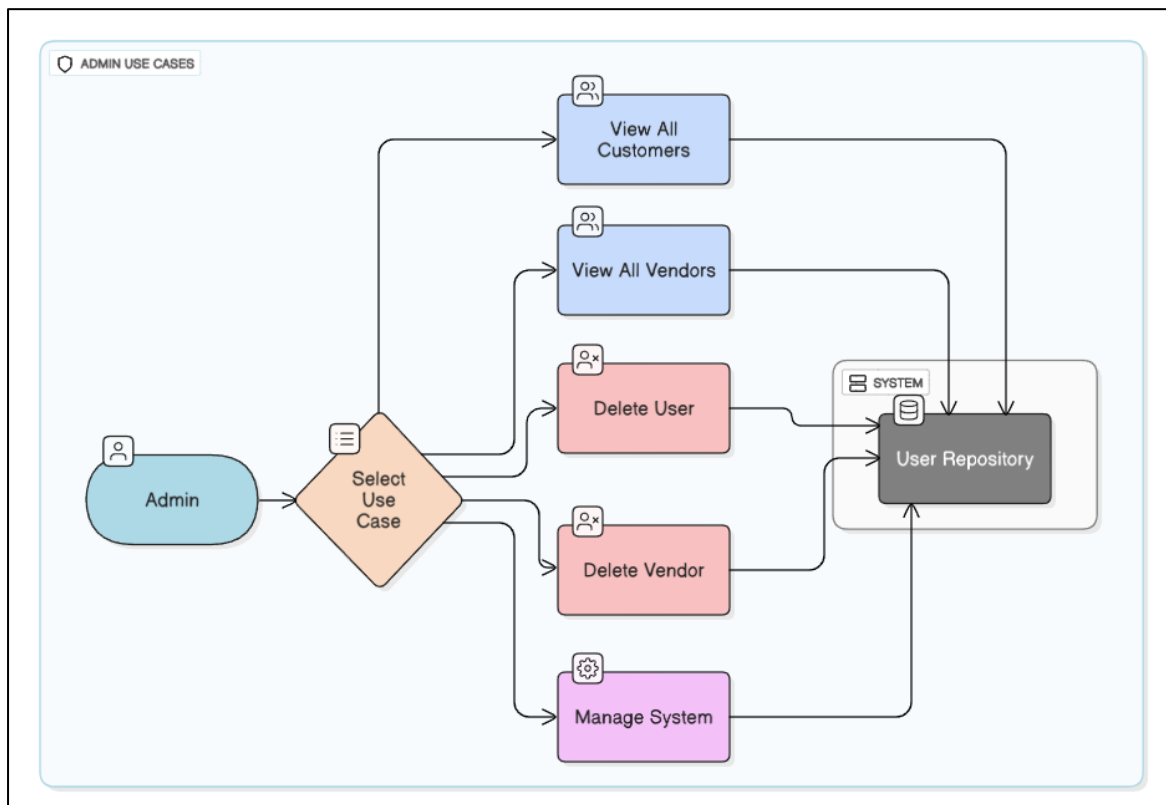
- **Use Case Diagrams:** Admin, Vendor, Customer.



Use Case Diagram of Customer Activities

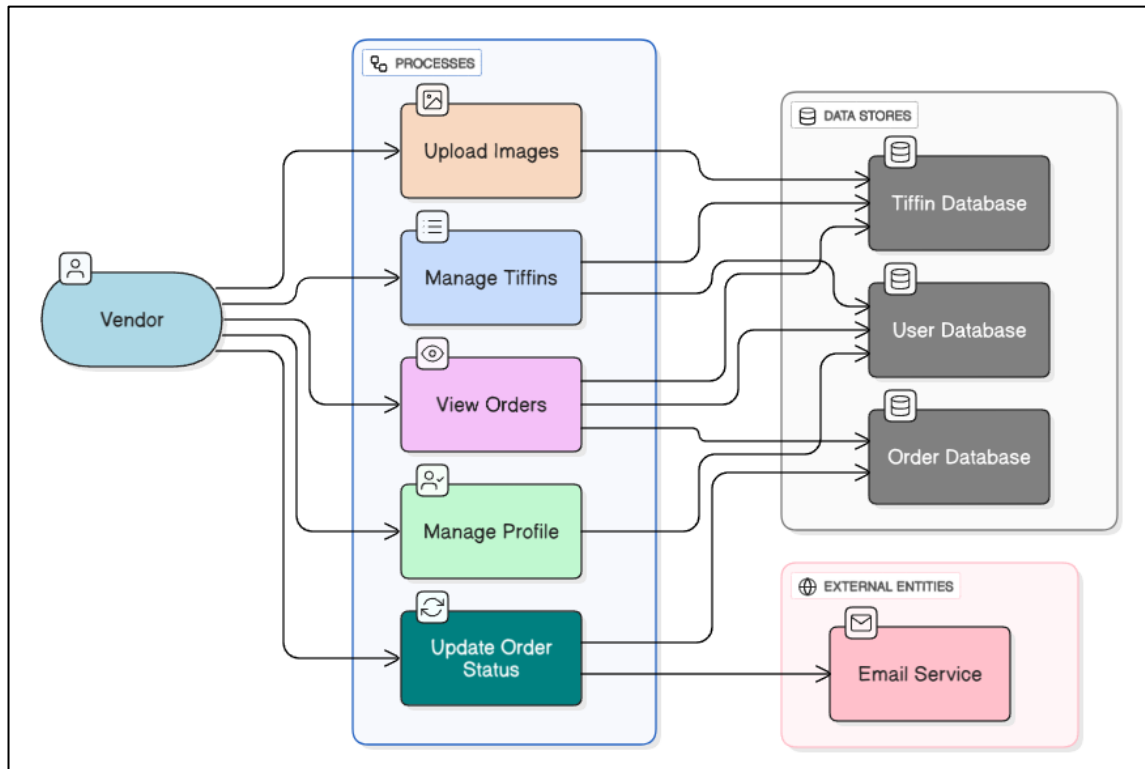


Use Case Diagram of Vendor Activities

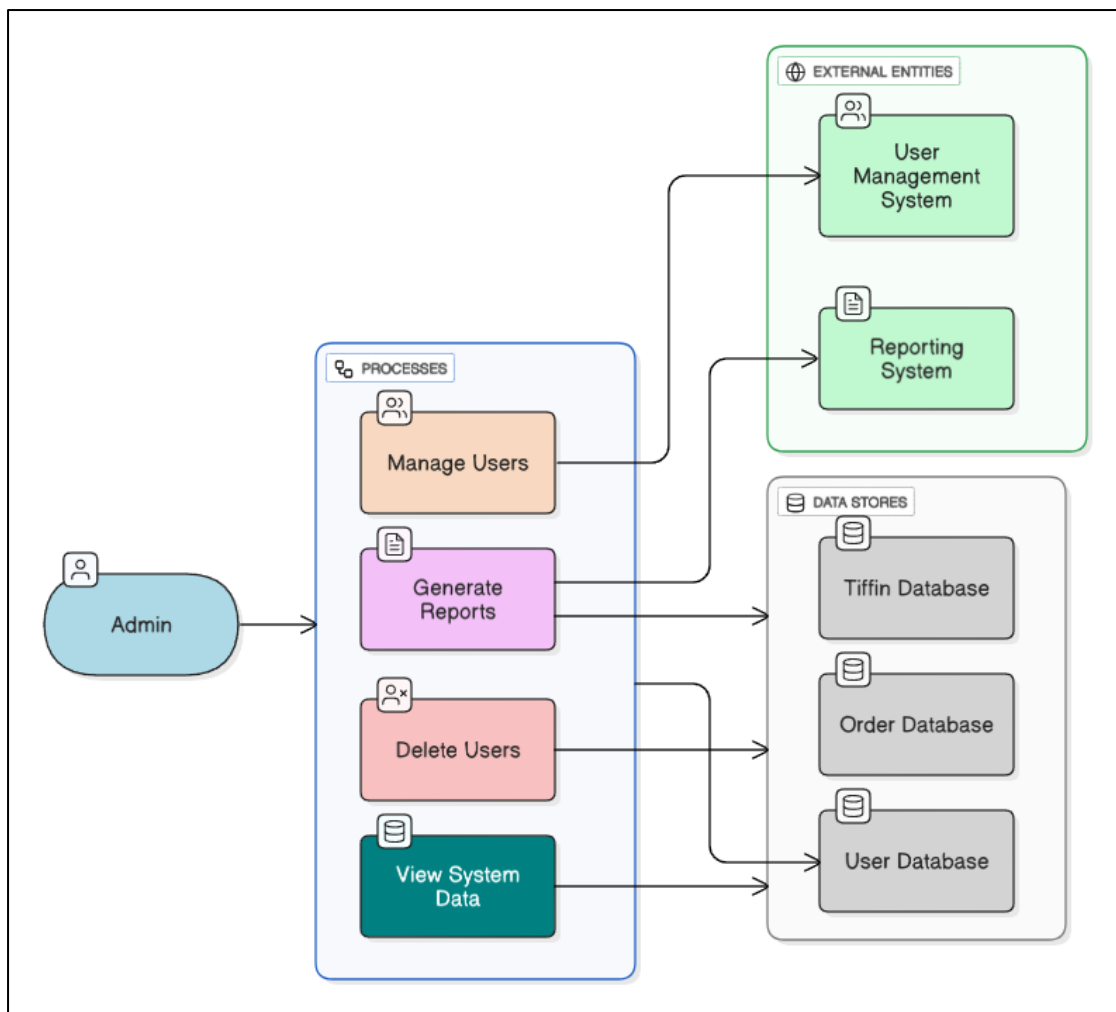


Use Case Diagram of Admin Activities

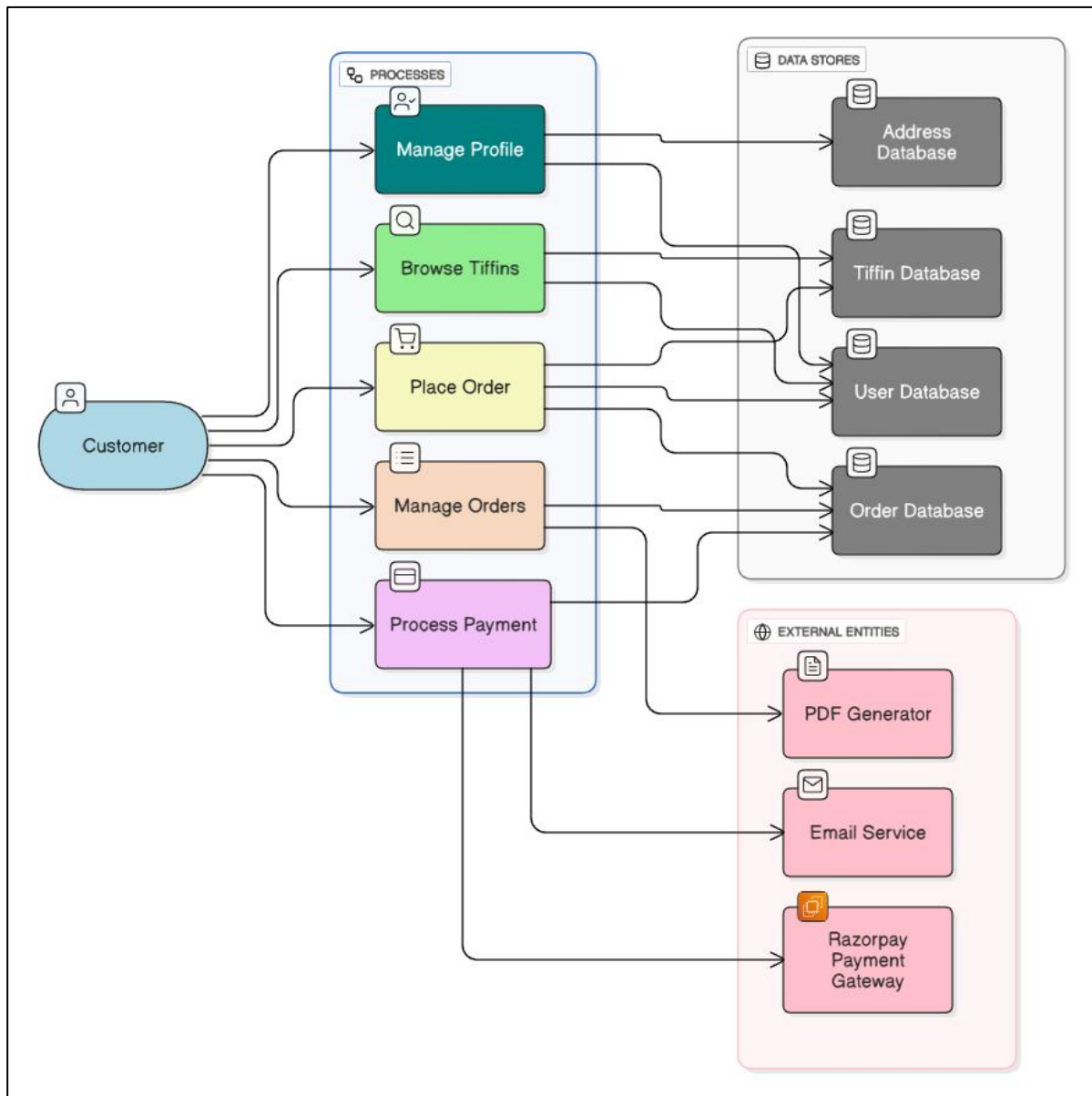
- **Level-1 DFDs:** Vendor, Customer, Admin.



Level 1 DFD (Vendor)

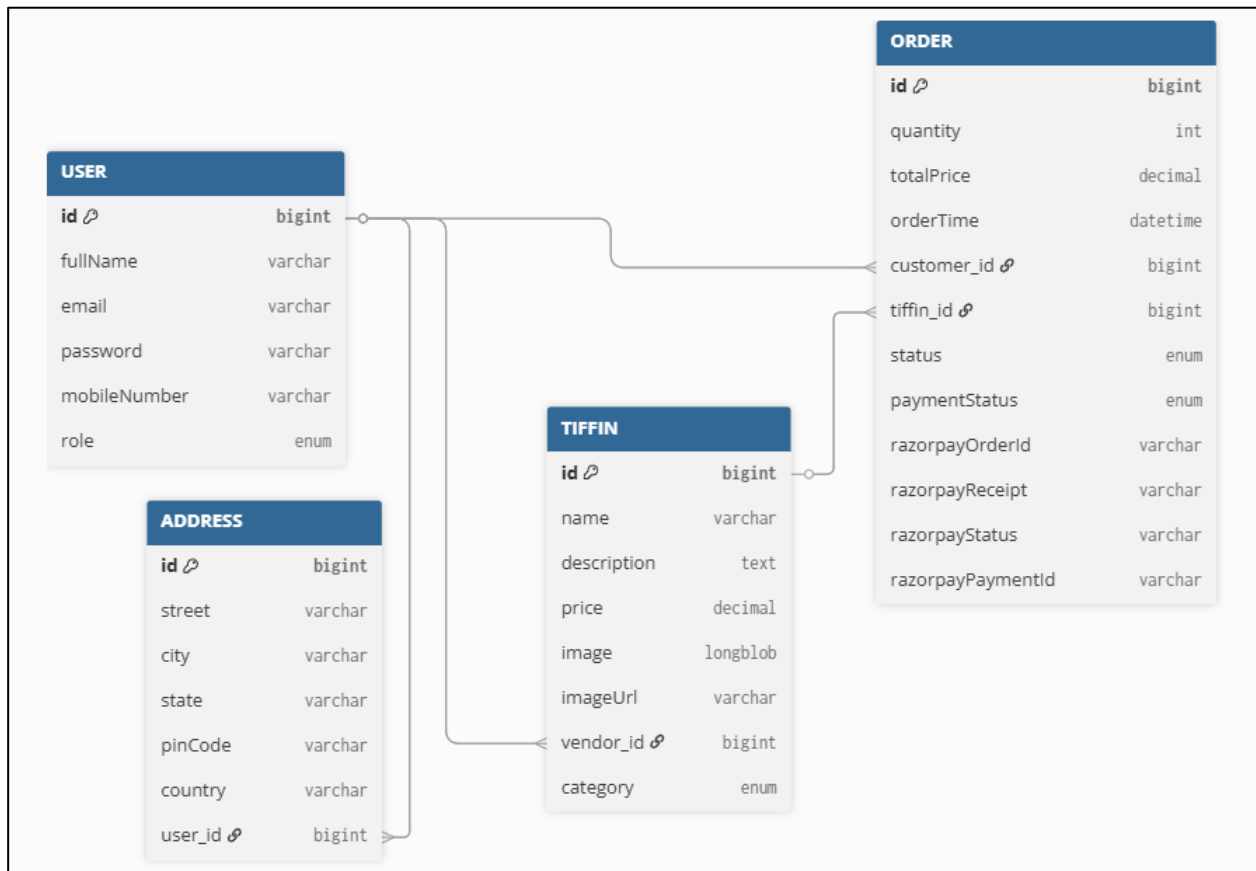


Level 1 DFD (Admin)



Level 1 DFD (Customer)

- **ER Diagram:** MySQL auto-generated.



ER Diagram